

Gabriel Torres

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EDUCATION

University of Costa Rica

Bachelor's Degree in Electrical Engineering, Honors

San Pedro, San José

Mar. 2019 – Feb. 2024

University of Costa Rica

Computer Science Coursework

San Pedro, San José

Mar. 2025 – 2026

PROJECTS

IMX219 Camera Driver Integration | *Embedded Linux, C, I2C, MIPI CSI-2, GStreamer*

- Integrated the Sony IMX219 MIPI CSI-2 camera sensor into an NVIDIA Jetson Nano running Embedded Linux
- Configured and validated Linux device tree definitions, I2C communication, and camera driver initialization
- Built Linux kernel modules and debugged sensor integration issues using kernel logs and hardware diagnostics
- Developed and tested GStreamer pipelines for camera capture and real-time video validation

Jetson Nano Embedded Linux Development | *Linux Kernel, ARM64, GCC, Docker, Yocto*

- Performed board bring-up and system flashing on the NVIDIA Jetson Nano using NVIDIA SDK Manager
- Cross-compiled and customized the Linux kernel, kernel modules, and device tree binaries using the Linaro GCC toolchain
- Configured and debugged embedded Linux environments using UART serial communication, kernel logs, and Linux command-line tools
- Explored Yocto-based embedded Linux development and Docker-based development environments

MIPI Camera Depth Estimation Application | *C++, GStreamer, Jetson Inference, H.264*

- Developed a C++ application for monocular depth estimation using camera input on the NVIDIA Jetson Nano
- Integrated GStreamer pipelines for real-time camera acquisition and video processing
- Implemented video streaming over UDP using hardware-accelerated H.264 encoding

EXPERIENCE

Project Engineer I

Feb. 2024 – Jul. 2025

Sistemas IQ, SIQ S.A.

Uruca, San José

- Designed and implemented control software for industrial automation systems
- Integrated software and hardware components according to technical and functional requirements
- Performed system commissioning, testing, and debugging to identify and resolve integration issues
- Developed validation procedures and technical documentation to support reliable system operation

Project Engineer Assistant

Oct. 2023 – Feb. 2024

Sistemas IQ, SIQ S.A.

Uruca, San José

- Developed PLC control logic and HMI interfaces for industrial automation applications
- Collaborated with engineering teams on requirements analysis, system integration, and troubleshooting

Undergraduate Research Assistant

Aug. 2022 – Jan. 2023

University of Costa Rica

ARCOSLab, INII

- Developed and validated an embedded force-sensing prototype for robotic applications using an STM32 microcontroller and C

TECHNICAL SKILLS

Programming: C++, C, Python, Java, Bash

Embedded Systems: Embedded Linux, Linux Kernel, Device Tree, Kernel Modules, I2C, MIPI CSI-2, ARM64

Multimedia: GStreamer, H.264, UDP Streaming, Camera Integration, Jetson Inference

Development Tools: Git, Docker, GCC, GDB, Make, CMake, Cross-Compilation

Platforms: NVIDIA Jetson Nano, Linux, Yocto (Familiarity)

Software Engineering: Object-Oriented Programming, Data Structures, Algorithms, Testing, Debugging

Languages: Spanish (Native), English (Advanced)